Great Learning x MIT Education Online June 2026

Great Learning Massachusetts Institute of Technology

CUSTOMER SEGMENTATION

UNSUPERVISED LEARNING

Applied Artificial Intelligence and Data Science Program

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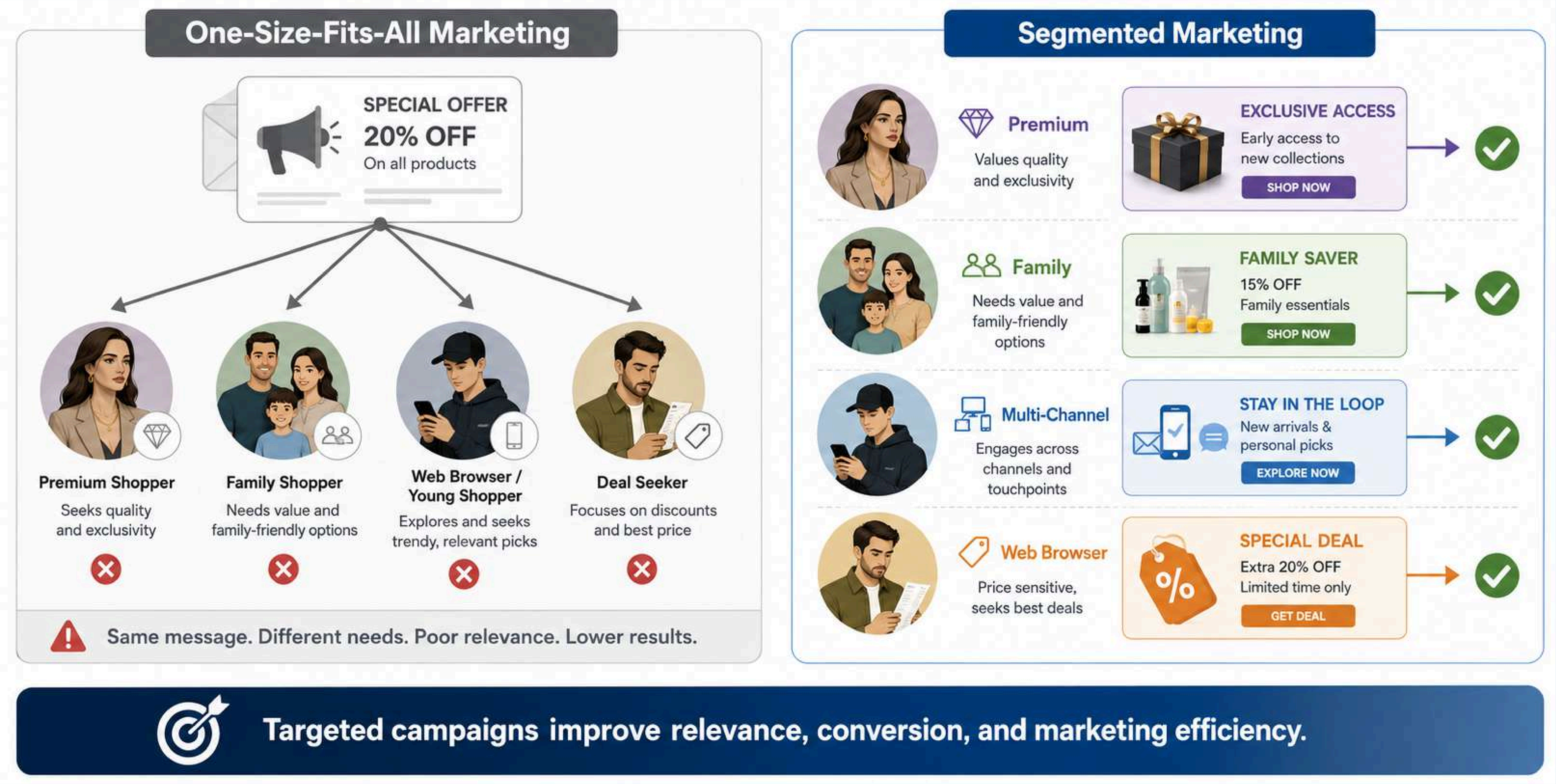
BUSINESS PROBLEM

Why one-size-fits-all Marketing is inefficient

- Customers differ in income, family structure, spending behavior, purchase channels, and campaign responsiveness.
- A single marketing strategy may waste budget by targeting low-response customers with the wrong offers.
- High-value customers may need premium loyalty offers, while budget-conscious customers may respond better to discounts or bundles.
- Website visits alone do not guarantee customer value; some customers browse often but spend very little.
- Customer segmentation helps the business allocate marketing resources more efficiently and personalize campaigns by customer group.

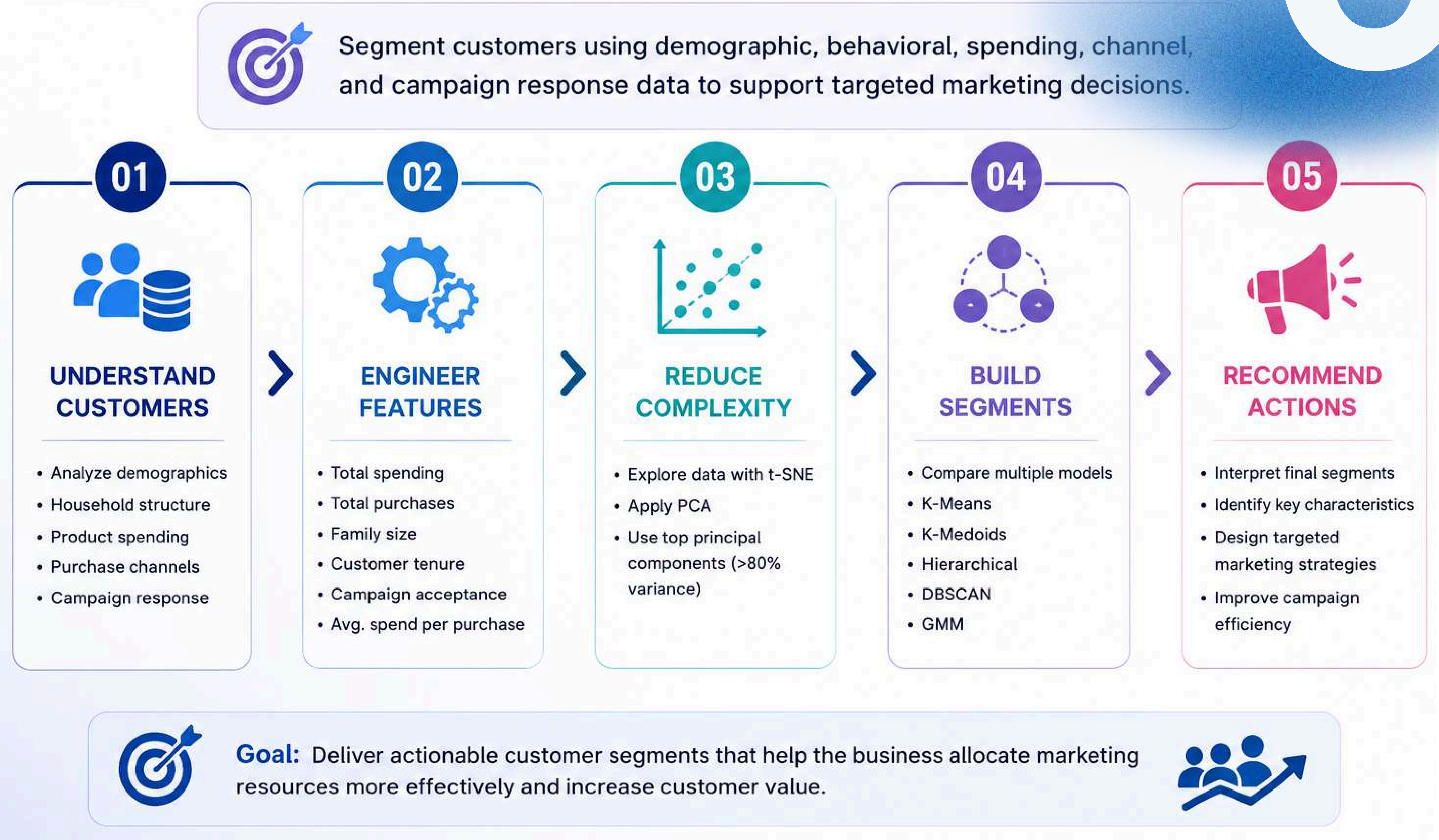
01

BUSINESS PROBLEM (CONT)

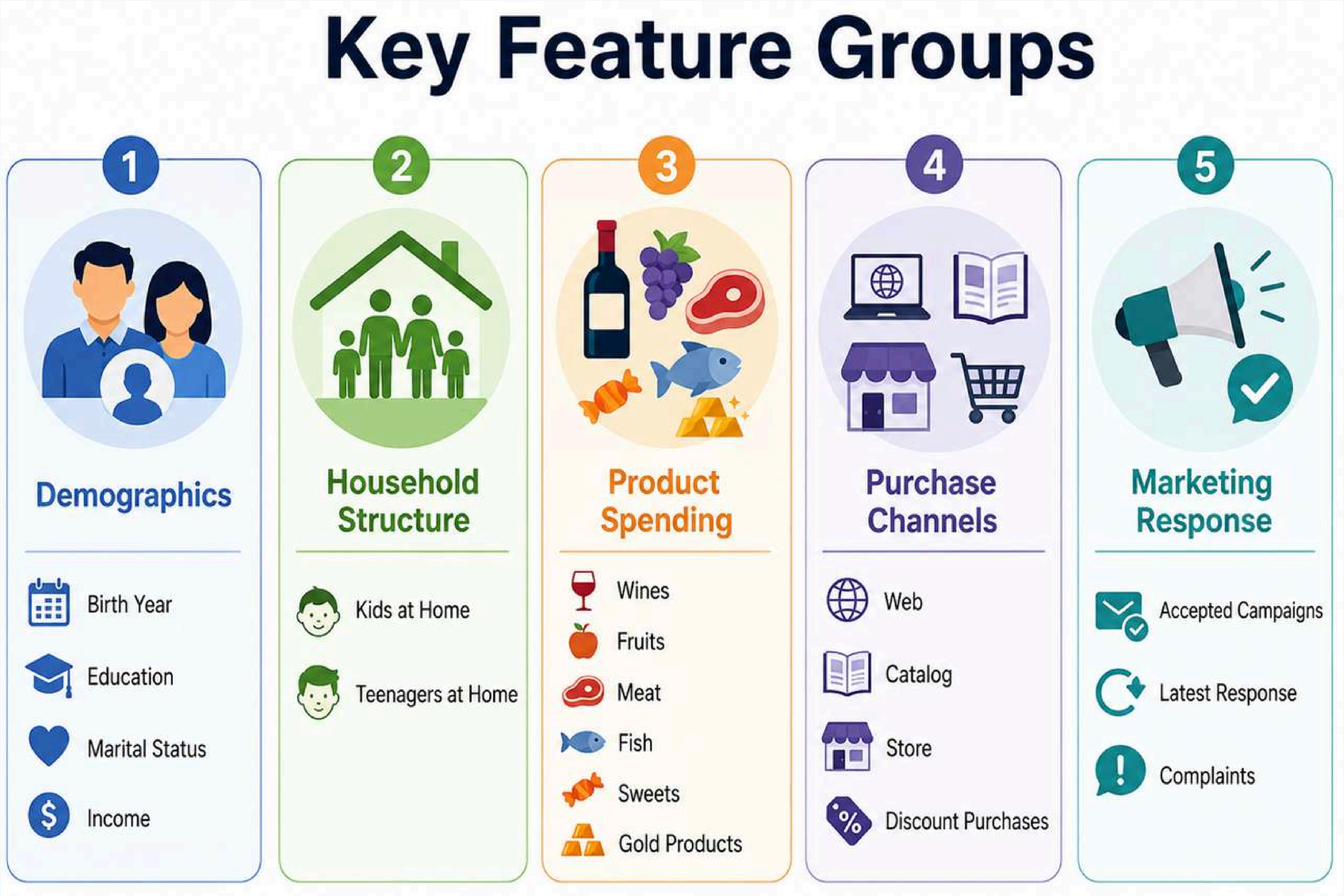


PROJECT OBJECTIVE

02



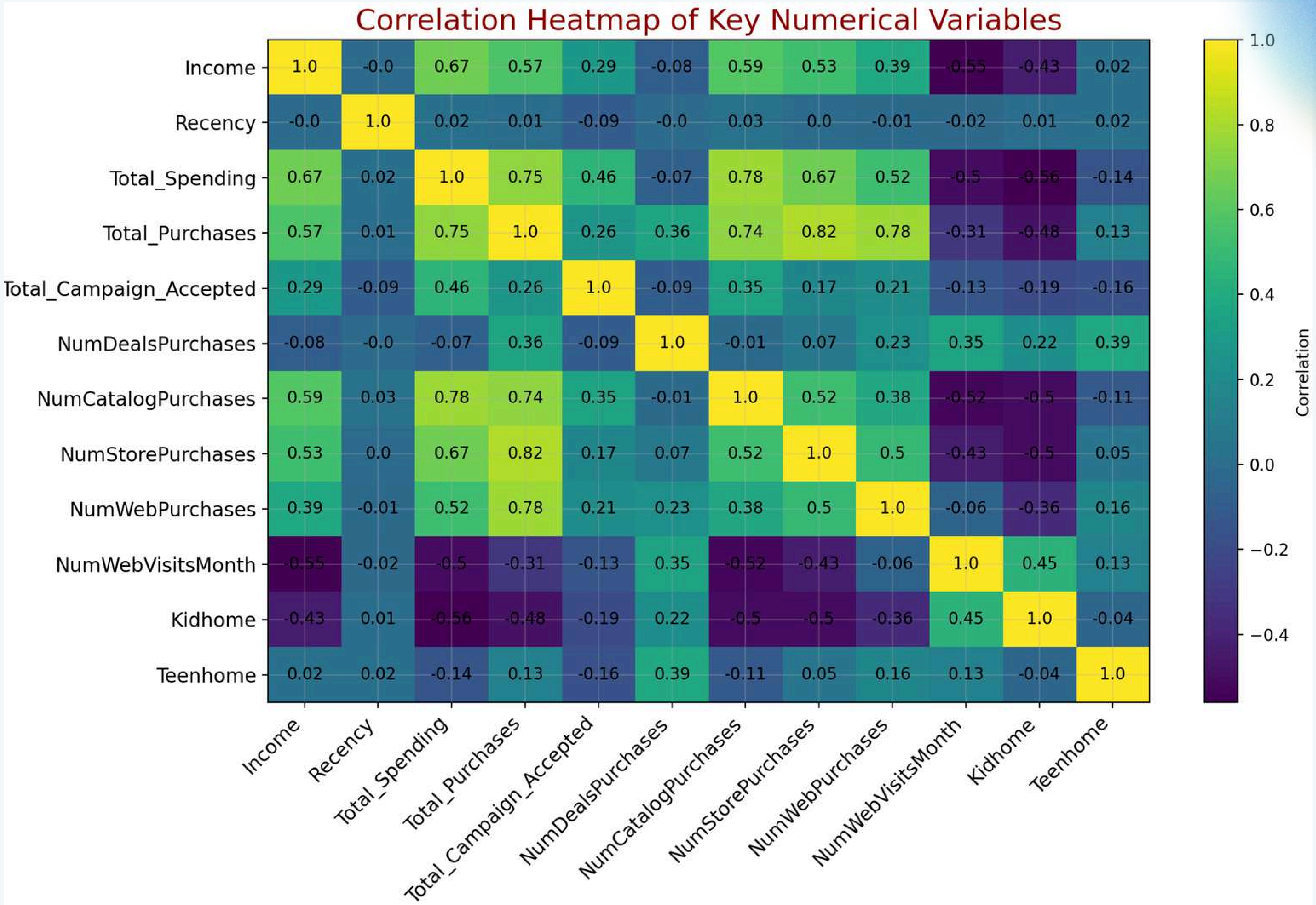
DATASET OVERVIEW



- The dataset contains **2,240 customer records** and **27 original variables**.
- Each row represents one customer and includes demographic, household, spending, purchase, and campaign response information.

03

EDA KEY INSIGHTS



EDA KEY INSIGHTS (CONT)

- **Income** and **spending** are **strongly related**: higher-income customers generally spend more and purchase more frequently.
- **Catalog** and **store purchases** are linked to **higher customer value**, while website visits alone are not enough to identify valuable customers.
- High **web visits** can indicate browsing, **not buying**: some customers visit the website often but have low spending and weak conversion.
- Customers with **more children** tend to **spend less**, suggesting larger households may be more budget-conscious.
- These insights support the need for segmentation based on both **customer value** and **purchase behavior**.

FEATURE ENGINEERING

05

Turning raw data into customer-level signals

Raw Data	Engineered Feature	Meaning
Year_Birth	Age	Customer life stage
Product spending columns	Total_Spending	Overall customer spending value
Purchase channel columns	Total_Purchases	Total buying activity across channels
Kidhome + Teenhome	Total_Children	Number of children in household
Household information	Family_Size	Size of the customer household
Dt_Customer	Customer_Tenure	Length of relationship with the company
Campaign response columns	Total_Campaign_Accepted	Overall responsiveness to marketing campaigns
Total_Spending + Total_Purchases	Amount_Spent_Per_Purchase	Average basket value per purchase

DATA PREPARATION

06

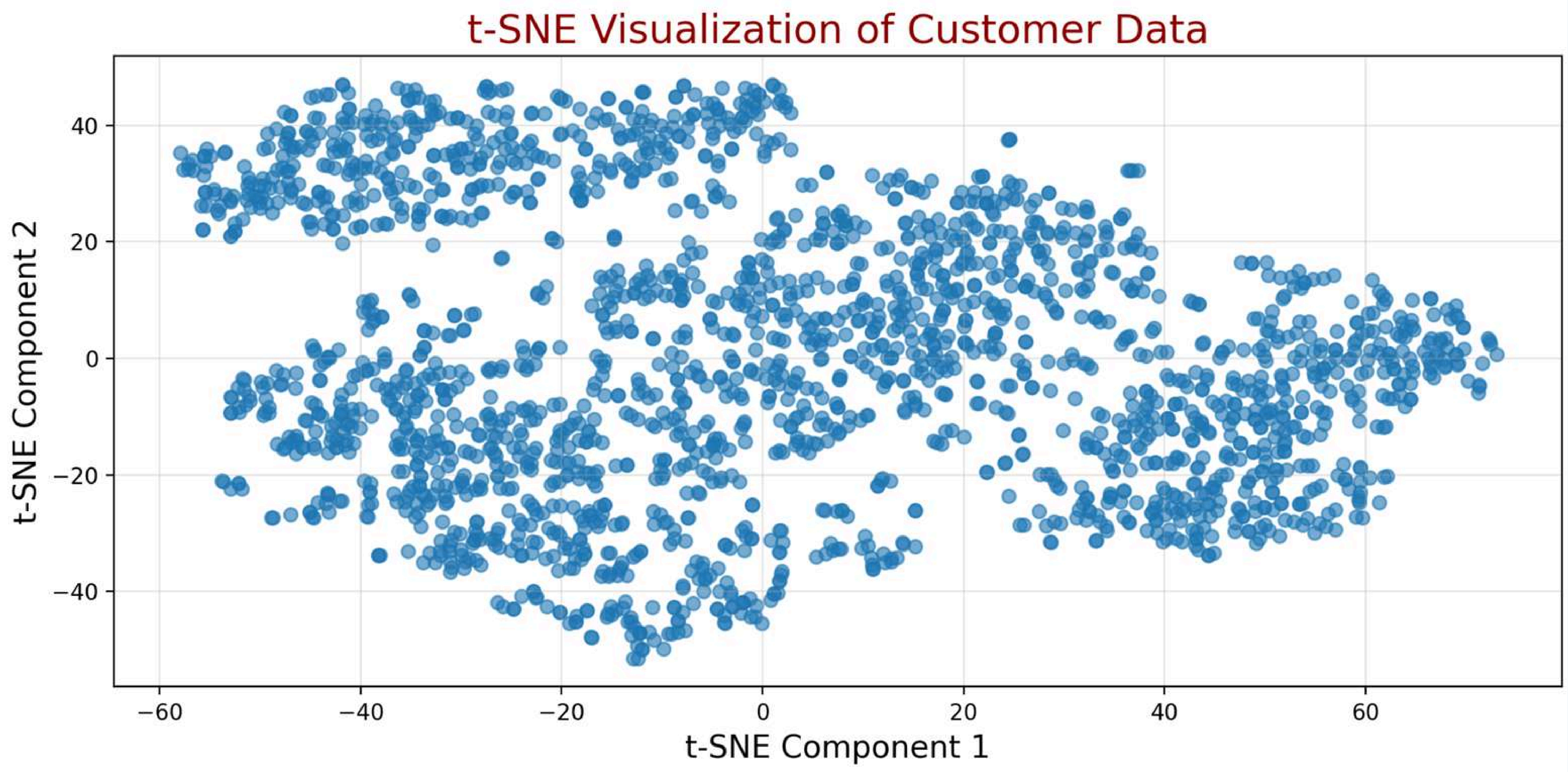
From raw features to segmentation-ready data

- Separated variables into **segmentation attributes** and **profiling attributes**.
- **Segmentation attributes** were used to create clusters, mainly behavior and customer-value variables: **income, spending, purchases, web visits, family size**.
- **Profiling attributes** were used after clustering to explain and interpret each segment: **education, marital status, product categories, campaign response**.
- **Removed irrelevant variables** before clustering: ID, raw dates.
- **Scaled numerical variables** so that features with larger values, such as income or spending, did not dominate the model.
- Applied **PCA** to reduce dimensionality and prepare a cleaner dataset for clustering.

DIMENSIONALITY REDUCTION

t-SNE

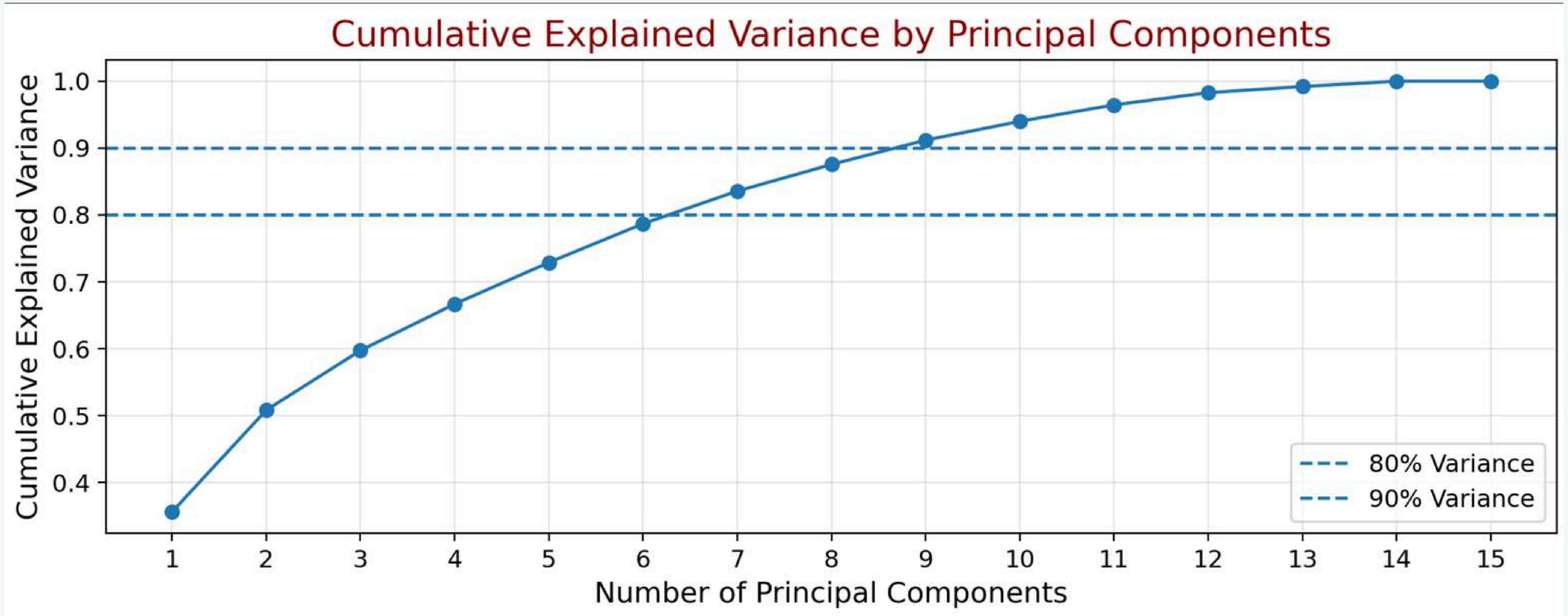
- Used t-SNE to visualize customer similarity in two dimensions.
- t-SNE showed some local customer groupings, but the groups were not perfectly separated.



DIMENSIONALITY REDUCTION (CONT)

PCA

- Used PCA to reduce dimensionality and summarize the main sources of variation.
- PCA helped reduce noise and handle correlation among variables before clustering.
- The first 7 principal components explained more than 80% of total variance.
- The first 9 principal components explained more than 90% of total variance.
- Clustering models were applied on the PCA-transformed dataset



MODEL TESTED

08

Tested 5 unsupervised clustering models to identify customer segments.

- **K-Means**: groups customers around cluster centroids; **simple and highly interpretable**.
- **K-Medoids**: similar to K-Means but **more robust to outliers** because it uses real observations as centers.
- **Hierarchical Clustering**: creates a dendrogram to explore **nested customer group structures**.
- **DBSCAN**: identifies high-density customer groups and noise/outliers.
- **Gaussian Mixture Model**: assigns customers **probabilistically** and allows more flexible cluster shapes.

➔ **Final selection was based on metrics + business usefulness, not metrics alone.**

MODEL COMPARISON

Multiple clustering techniques were compared using **both technical metrics and business interpretability.**

- K-Means at K = 2 had the highest silhouette score, but the result was **too broad for marketing strategy**.
- K-Means at K = 4 provided more **actionable customer segments while still maintaining reasonable performance**.
- K-Medoids produced similar insights but **did not significantly improve over K-Means**.
- Hierarchical Clustering created one very small **outlier-like cluster**, making it less practical.
- DBSCAN produced **highly imbalanced clusters**, with most customers grouped into one large cluster.
- GMM produced balanced clusters, but had **lower silhouette performance and was more complex to explain**.

Final choice: K-Means with 4 clusters, because it was interpretable, balanced, simple, and marketing-actionable.

MODEL COMPARISON (CONT)

Model	Main Result	Decision
K-Means	Interpretable 4-cluster solution	Selected
K-Medoids	Similar to K-Means	Supporting model
Hierarchical	Tiny outlier-like cluster	Not selected
DBSCAN	Severe imbalance	Not selected
GMM	Balanced but lower silhouette	Supporting model

Final choice: K-Means with 4 clusters —
best balance of interpretability, cluster balance, simplicity, and marketing actionability.

FINAL CUSTOMER SEGMENTATION

K-Means 4-Cluster Solution: Segments name chosen for business interpretation

RECOMMENDATIONS & NEXT STEPS

Implementation Roadmap

1 Assign Segment Labels

- Apply the K-Means 4-cluster labels to all customers
- Store the segment ID and segment name in the CRM/database

2 Launch Targeted Campaigns

- Design one campaign strategy for each segment
- Examples: loyalty offers, bundles, retargeting, conversion incentives

3 Monitor KPIs

- Track response rate, conversion rate, AOV, revenue, and retention
- Measure performance by customer segment

4 A/B Test Campaigns

- Compare segmented campaigns vs generic campaigns
- Test offers, channels, and messaging

5 Refresh the Model

- Re-run segmentation every 3–6 months
- Update features as customer behavior changes

Future improvements

- Customer Lifetime Value
- Product Preference Ratios
- Campaign Response Prediction
- New Transaction Data

12 THE JOURNEY CONTINUES

Limitations and future improvements

- This is an academic capstone project, so the dataset is smaller and cleaner than large-scale real business data. The project should be viewed as a foundation and reusable template for customer segmentation, not a final enterprise-level production system.
- More customer data could improve the model, including transaction history, browsing behavior, product returns, loyalty activity, and customer lifetime value.
- Additional features could be engineered, such as product preference ratios, spending per family member, campaign response rate, and recency-frequency-monetary indicators.
- More models and advanced methods could be explored, including ensemble clustering, HDBSCAN, spectral clustering, and customer lifetime value modeling.
- Hyperparameter tuning could be improved using more systematic search methods instead of mainly testing selected parameter ranges manually.
- Marketing domain expertise is needed to validate whether the segments make sense commercially and whether the recommended campaigns are realistic.
- The segmentation should be updated over time because customer behavior, preferences, and campaign response patterns can change.

THANK YOU FOR YOUR ATTENTION

Github Repository:

https://github.com/noahnguyen296/statistics-machine-learning-projects/tree/main/customer-segmentation-unsupervised-learning

Short presentation video:

https://www.youtube.com/watch?v=JMAi2kLO250

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